

Joshua Smith

BScEng Electrical & Computer Engineering · University of Cape Town

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PROFILE

Final-year ECE student with a strong foundation in software engineering, algorithms, and systems programming. Comfortable working across platforms, from low-level embedded C to real-time processing pipelines, and driven by problems where performance, correctness, and real-world behaviour under load matter. Highly collaborative and focused on producing clean, scalable code for complex engineering challenges.

EDUCATION

BSc(Eng) Electrical & Computer Engineering *University of Cape Town, Cape Town* 2023 – Present

- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Embedded Systems (C), Computer Architecture, Linear Algebra, Control Systems, Digital Signal Processing

TECHNICAL SKILLS

Languages: Python, C, MATLAB, Bash

Computer Fundamentals: Data Structures, Algorithms, OOP, Concurrency and Parallelism, Systems programming, Complexity analysis

Tools & OS: Git/GitHub, Linux (Kali / Raspberry Pi), VS Code, Simulink, StateFlow

Domains: Real-time processing, Edge inference, Embedded software, Control systems, High-Performance Computing

PROJECTS

StarCore-1 RISC Processor & CRU Integration – iVerilog, MATLAB, Computer Architecture

- Designed a central processing unit based on the RISC architecture and ISA, extending the baseline StarCore-1 datapath by integrating a custom Coordinate Rotation Unit (CRU)
- Developed a high-precision CORDIC algorithm simulation in MATLAB to generate golden test vectors and rigorously validate hardware accelerator correctness
- Simulated full datapath correctness and ISA integration via the creation of custom testbenches in iVerilog

Medical Imaging & Convolutional Neural Networks – Python, MATLAB

- Developed mathematical scripts to execute advanced image processing fundamentals, including computing Radon transforms, running X-ray attenuation simulations, and implementing image registration
- Implemented, trained, and evaluated Convolutional Neural Networks (CNNs) to accurately identify and classify pneumonia in patient X-ray datasets

Embedded Control System – C, STM32, real-time design

- Designed and validated a USB-CPD-based Li-ion charging system: state machine design, voltage negotiation logic, and protection behaviour verified through simulation and hardware testing

EXPERIENCE

Engineering Start Up Intern *World Optimum Engineering, Cape Town*

Jul 2026 – Present

- Currently engineering systems to seamlessly combine additive manufacturing with 3D scanning technologies to accelerate prototype development and evaluation

LEADERSHIP & EXTRACURRICULARS

Prescient Coding Challenge – Algorithm & Software Engineering Competitor

- Preparing to compete in the upcoming September 2026 algorithmic and software engineering competition alongside computer science peers

Academic Mentor & Tutor – University of Cape Town

- Volunteer mentor for first-year university engineering students and part-time tutor for high school students, focusing on breaking down complex mathematical and foundational engineering concepts

UCT Developers' Society – Active Member

- Active participant in the university software development club, consistently engaging with peers on modern software engineering practices and collaborative technical problem-solving